



Grade 3 Science Booklets (Ontario Curriculum Aligned)

Below are outlines for **five** Grade 3 science booklets, each aligned with the Ontario Science & Technology curriculum (TVO Learn). Four booklets cover the major science strands (Life Systems, Structures & Mechanisms, Matter & Energy, Earth & Space Systems), and the fifth booklet is a standalone “Just the Facts” reference. Each booklet is designed to be **print-friendly (A4, black-and-white)** and includes all required sections (Title Page, Table of Contents, content sections, diagrams placeholders, review questions, and glossary).

Booklet 1: Understanding Life Systems – Growth and Changes in Plants

Title Page & Parent Note: *Growth and Changes in Plants*

Note to Parents/Caregivers: This booklet supports your child’s learning about plants in Grade 3 Science. It aligns with the Ontario curriculum strand “**Understanding Life Systems: Growth and Changes in Plants**” ¹. Inside, you’ll find key concepts about what plants need, the parts of plants, how plants grow and change, and how they make food. Each section includes simple explanations and a few diagrams. Encourage your child to discuss the review questions with you. Enjoy exploring the amazing world of plants together!

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(Note: Page numbers are approximate placeholders.)

Section 1: What Do Plants Need?

Plants are living things, and like all living things they have basic needs for survival. **Plants need sunlight, water, air, and nutrients from soil to live and grow** ² ³. Sunlight provides energy, water and nutrients help them build their tissues, and air (particularly carbon dioxide) is used to make food. If a plant is missing any of these essentials, it may not grow properly. For example, without water a plant will dry out, and without light a plant cannot make food and will turn pale or stop growing. This section explores each of these needs and how plants get them in their environment.

(Diagram: A young plant in soil with labels for sunlight, water, air, and nutrients showing what the plant needs.)

Most plants get **water and nutrients** from the soil through their roots. They also need **air**; in fact, plants “breathe” carbon dioxide from air and release oxygen. **Sunlight** is crucial because it is the energy source for plants. In later sections, we will see how plants use sunlight to make food. Different plants may need these things in different amounts (for instance, cacti need less water than water lilies), but **all plants require water, air, nutrients, and light to survive** ² ⁴ . Knowing what plants need helps us take care of them in gardens or at home.

Section 2: Parts of a Plant and Their Functions

Plants might stay in one place, but they have specialized parts that help them live and grow. The **major parts of most plants are the roots, stem, leaves, flowers, and fruits (which contain seeds)** ⁵ . Each part has an important job (function):

- **Roots** grow underground and anchor the plant. Roots absorb water and nutrients from the soil, acting like the plant’s drinking straws. They also store food for the plant in some cases (like carrots, which are actually a root we eat!).
- **Stem** (or trunk in a tree) is the main support that holds up the plant. The stem carries water and nutrients from the roots to the leaves, and it also carries food (sugars) made by the leaves to other parts. A strong stem helps the plant stand upright.
- **Leaves** are like the plant’s food factories. They capture sunlight and air. Leaves are usually flat and green; the green color comes from chlorophyll, which helps in making food. Through a process we’ll discuss later, leaves use sunlight to turn water and air into sugars the plant can use ⁶ . Leaves also release oxygen into the air, which is important for us!
- **Flowers** are the reproductive part of many plants. They often have bright colors or scents. A flower can turn into a **fruit** after pollination. Inside the fruit, we find **seeds**. Fruits protect the seeds, and when fruits drop or are eaten and released by animals, the seeds can start new plants.
- **Seeds** contain baby plants (embryos) and a little food to help the new plant start growing. When a seed has the right conditions (water, warmth, air), it will sprout.

(Diagram: A flowering plant labeled with root, stem, leaf, flower, fruit, and seed.)

Each part of a plant works together to help the plant survive. For example, the roots take in water, the stem carries that water to the leaves, and the leaves use the water (and sunlight and air) to make food. If one part is damaged (like roots drying out or leaves being eaten), it can affect the whole plant’s health. By understanding plant parts, we can better care for plants (like knowing to water the soil for roots, or not to damage the stem so water can flow).

Section 3: How Do Plants Grow? (Life Cycle)

Plants grow and change through a **life cycle**. A typical life cycle of a flowering plant starts with a **seed**. When a seed is planted in soil and gets water and warmth, it begins to **germinate** – this means the seed coat breaks open and a tiny root and shoot emerge ⁷ . The root grows downward to find more water, and the shoot grows upward toward the light.

The shoot becomes a **seedling**, which is a baby plant with a few small leaves. As it gets more sunlight, water, and nutrients, the seedling grows bigger into a **young plant**. It grows more leaves (for making food) and a stronger stem. In time, many plants will develop **buds** that bloom into **flowers**. The flowers can then be pollinated (often with the help of insects or wind), and after pollination, the flowers develop into **fruits**. The fruit holds new seeds, and eventually the fruit falls or releases seeds, which can start the cycle again by sprouting into new plants ⁸.

Throughout the life cycle, plants undergo changes. For example, think of a sunflower: it starts as a seed, grows into a small green seedling, then into a tall plant with a big yellow flower. After the flower is pollinated, it forms a brown center filled with sunflower seeds – those seeds can be planted to grow more sunflowers. Other plants, like trees, have longer life cycles (a tree seedling may take years to become a big tree). But all plants share the pattern of **growth (getting bigger and producing leaves), reproduction (producing seeds), and eventually starting new life cycles**.

(Diagram: The life cycle of a plant, showing seed → seedling → mature plant → flower/fruit → seed again.)

It's important to note that not all plants grow from seeds; some grow from spores (like ferns) or cuttings. However, flowering plants and most of the plants we see daily do grow from seeds. The life cycle helps ensure that plants can continue to survive and multiply. By understanding life cycles, we learn why it's important, for instance, to leave wildflowers long enough to seed or how planting seeds in a garden yields new plants each year.

Section 4: How Plants Survive and Adapt

Plants cannot move around like animals, but they have amazing ways to survive in their environments. An **adaptation** is a special feature or behavior that helps a living thing survive where it lives. Plants have many adaptations to handle different conditions.

For example, **desert cacti** have thick, fleshy stems that store water, and spines instead of broad leaves (spines lose less water and protect the cactus from animals). **Water lilies** have wide flat leaves that float on water, which helps them get sunlight while their roots stay in muddy water. Some trees in cold climates are **evergreen conifers** with needle-like leaves; these needles have a waxy coating to resist freezing and water loss, allowing the tree to survive winter.

Another important survival strategy for plants is **seed dispersal** – getting their seeds spread out so they can grow in new places. Different plants disperse seeds in different ways. **Maple trees**, for instance, have winged seeds (“helicopter” seeds) that spin and travel by wind ⁹. **Burdock plants** have burrs that stick to animal fur (or your clothes!) and get carried away. **Coconut palms** drop coconuts that float in water to new islands. These are all adaptations to help the plant's offspring (new plants) find a good spot to grow.

Plants also adapt by responding to their environment as they grow. Have you seen a houseplant bend toward a window? That's an adaptation called **phototropism** – the plant grows toward the light to get more energy. Roots often grow toward moisture in the soil. Some vines can sense support structures and wrap around them to climb higher. These behaviors help the plant get what it needs (light, water, support).

Through adaptations, plants have become capable of living almost everywhere on Earth – from hot, dry deserts to deep underwater, and from rocky mountain tops to shaded forests. Each adaptation increases a

plant's chance of survival in its particular habitat. In this section, we explore several cool plant adaptations and how they work.

Section 5: Plants and Sunlight: Introduction to Photosynthesis

One of the most remarkable things about plants is that they **make their own food**. They do this through a process called **photosynthesis**. The word “photosynthesis” literally means “putting together with light” ¹⁰. In photosynthesis, green plants use **energy from the Sun** to transform carbon dioxide (a gas in the air) and water into **sugar (food)** and **oxygen** ⁶. This happens mainly in the leaves, which contain chlorophyll (the green pigment that captures light).

In simple terms, the equation of photosynthesis is:

Sunlight + Water + Carbon Dioxide → Sugar (food) + Oxygen.

During daytime, a plant's leaves take in sunlight and carbon dioxide. The roots bring water to the leaves. Inside the leaves, using the Sun's energy, the plant changes the water and carbon dioxide into a type of sugar (glucose) that it uses for food. Oxygen is produced as a **by-product** (extra output) and released into the air. **This is great for us, because we breathe the oxygen that plants make!** ¹¹ In fact, a large portion of the oxygen in Earth's atmosphere is made by plants and algae. At night or when there's no sun, photosynthesis stops, but the plant will use the food it made to keep living and growing.

Photosynthesis is crucial not only for the plant's own growth but for life on Earth in general. It's how energy from the Sun gets into the food chain. When we eat plants (or animals that ate plants), we're using the energy the plants stored. Also, photosynthesis is why we have breathable oxygen. In this section, we introduce photosynthesis in a Grade-3-friendly way, with diagrams to illustrate how a leaf works. Even though the full details of photosynthesis are complex, the key idea is: **plants use sunlight to make food** ³.

(Diagram: A simple diagram of a leaf in sunlight, showing arrows for sunlight, CO₂ going in, O₂ going out, and sugar being produced within the leaf.)

Understanding photosynthesis also helps explain why plants need light and why they have green leaves. It's the reason you should put a plant near a window – without light, a plant can't make food and will eventually die. Plants truly are nature's solar-powered food factories!

Review Questions (Understanding Life Systems: Plants)

1. **List three things that plants need to grow.**
2. **What part of a plant absorbs water and nutrients from soil?**
3. **How do plants spread their seeds?** (Name one method of seed dispersal.)
4. **What is photosynthesis?** (Explain in a simple sentence.)
5. **Why are leaves important to a plant's survival?**

These questions cover the key ideas from the booklet. Encourage students to answer in their own words. For example, for question 4, a good answer might be: “Photosynthesis is the process by which plants use sunlight to make their own food and produce oxygen.” Reviewing these questions helps reinforce

understanding of plant biology and ensures the student can recall important facts (like basic needs, parts of plants, and the concept of photosynthesis).

Glossary (Plants Unit)

- **Adaptation:** A special feature or behavior that helps a living thing survive in its environment (e.g. a cactus's thick stem is an adaptation to store water).
- **Flower:** The part of a plant that can produce seeds (often colorful and scented to attract pollinators). After pollination, a flower can turn into a fruit containing seeds.
- **Germination:** The process by which a seed begins to grow and puts out roots and a shoot ⁷. Germination is the first step in a plant's life cycle, when a seed "wakes up" and starts growing into a seedling.
- **Nutrients:** Substances that living things need for growth and energy. Plants absorb nutrients like minerals from the soil through their roots.
- **Photosynthesis:** The process by which green plants use sunlight, water, and carbon dioxide to make their own food (sugars) and oxygen ⁶. This happens in the leaves using a green pigment called chlorophyll.
- **Root:** The part of a plant that usually grows underground. Roots hold the plant in place and take up water and nutrients from the soil.
- **Seed:** The small part of a plant that can grow into a new plant. It contains a baby plant (embryo) and a food supply inside a protective coat.
- **Seedling:** A young, small plant that has just grown from a seed. It usually has a short stem and a few leaves.
- **Sunlight:** The light from the Sun, which provides energy. Plants need sunlight to perform photosynthesis and make food.
- **Transpiration:** (Optional advanced term) The process by which water evaporates from a plant's leaves. It helps pull water up from the roots. *(Included if teacher wants vocabulary extension.)*

(This glossary can be expanded or trimmed as needed. It covers key terms in the Grade 3 "Growth and Changes in Plants" topic. Students can refer to it for spelling and understanding while reading the booklet.)

Booklet 2: Understanding Structures and Mechanisms – Strong and Stable Structures

Title Page & Parent Note: *Strong and Stable Structures*

Note to Parents/Caregivers: This booklet aligns with the Ontario Grade 3 strand **"Understanding Structures and Mechanisms: Strong and Stable Structures."** It introduces students to what structures are and how we make them strong and stable. Your child will learn through examples of everyday structures (bridges, buildings) and natural structures (animal homes). The sections include simple explanations of terms like **strength, stability, and balance**, with diagram placeholders to visualize concepts. Use the review questions to talk about structures around you (like asking: why does a chair have four legs?). This will help your child connect science to the real world.

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Section 1: What Is a Structure?

In science, a **structure** is any object with a definite shape and size that is built or formed to serve a purpose. A structure is essentially a **supporting framework that holds a load** ¹². This means structures can support weight or resist forces without collapsing. Structures can be **natural** (made by nature) or **human-made** (built by people).

Examples of natural structures: A tree is a structure – its trunk is like a tower holding up branches and leaves. A spider’s web is a delicate structure designed to catch insects. A bird’s nest, a beaver’s dam, and even the human skeleton inside your body are all structures too! They have form (shape) and function (what they do).

Examples of human-made structures: Buildings (like houses, schools, skyscrapers), bridges, towers, dams, and even simple things like chairs and tables are structures. Each has a frame or shape that makes it useful: a chair’s structure supports the weight of a person, a bridge’s structure allows it to span a gap and hold vehicles, etc.

Every structure must handle **loads** (forces or weights) without breaking or deforming too much. For instance, a house’s walls and beams support the load of the roof and the people/furniture inside. If the structure is poorly designed or built from weak materials, it might fail when a load is placed on it. Engineers and even animals (like birds building nests) must consider how to make structures strong enough for their purpose.

In this section, we define structures and identify them in the world around us. Students are encouraged to point out structures at home or in nature and think about *what the structure’s purpose is* and *what kinds of loads or forces* it must withstand. For example: a playground slide’s structure supports children (the load) and stays firmly on the ground; a snail’s shell is a natural structure that protects the snail’s body.

Section 2: Strong Materials and Shapes

What makes a structure **strong**? A structure’s strength is its ability to support a load or resist forces without breaking ¹³ ¹⁴. Two key factors contribute to strength: the **materials** used and the **shapes/design** of the structure.

Materials: Some materials are stronger or tougher than others. For example, steel and concrete are very strong, which is why they are used to build skyscrapers and bridges. Wood is also strong (especially certain woods like oak) and is used in many house frames. Brick and stone are strong under compression

(squeezing force), which is why many foundations or walls are brick/stone. In contrast, materials like paper or glass are weaker (though they can have specific uses). **Using the right material for the job is important** – you wouldn't build a big bridge out of paper, and you wouldn't make a shirt out of concrete! In this section we'll compare materials and discuss concepts like hardness, flexibility, and durability as properties that affect strength.

Shapes and Design: Interestingly, the shape of parts of a structure can make it stronger **without changing the material**. For example, **triangles are known as one of the strongest shapes** for building ¹⁵. If you make a structure out of triangles (like a truss bridge or the triangular roof frames in houses), it holds its shape under heavy load and doesn't easily collapse. **Why triangles?** Because a triangle spreads force evenly through all three sides; it's hard to change its angles without changing the length of a side, so it's very rigid. In contrast, a square can tilt into a diamond shape if not braced, unless you add a diagonal (which creates two triangles). Engineers use triangles in bridges, towers (like power line pylons), and cranes for stability ¹⁶.

Other shape/design considerations include: **arches** (strong for bridges and doorways – they redirect weight to the sides and down), **columns** (vertical supports that hold weight from above), and **struts/ties** (diagonal braces in frameworks to resist bending). **Beams** are horizontal bars that support loads (like a floor or roof), and they need to be strong or supported to not break under bending forces.

We'll include simple activities or examples, like making a triangle with popsicle sticks vs. a square, to show how the triangle is more rigid. Students learn that both material and design work together – a strong material in a poor design can fail, and a great design made of weak material can also fail. The best structures use good materials and smart design.

(Diagram: A comparison of shapes – a square frame versus a triangle frame – with an arrow showing the square deforming into a diamond, but the triangle holding shape. Caption: "Triangles help structures stay rigid.")

Section 3: Stable Design (Staying Upright)

A strong structure isn't useful if it topples over. **Stability** is the ability of a structure to stay balanced and not fall when forces act on it ¹⁴. One principle of stability is having a **low center of gravity and a wide base**. **Structures are more stable when they have a wider base** ¹⁷. This is why a pyramid (wide at the bottom, narrow at top) is extremely stable, or why we spread our feet apart to balance better. If the weight of a structure is spread out low and wide, it's harder to tip it over.

Think of building blocks: a tower with a big base (like 4 blocks by 4 blocks at the bottom) can go taller without tipping than a tower that is just one block wide at the bottom. That's because the wide base supports it in all directions. Many real structures follow this idea: **buildings often have foundations wider than the upper floors, dams are built thicker at the bottom, and trees have wide-spreading roots or trunks that widen at the base**. Even simple furniture: a chair has four legs spread out at the corners, and often the legs flare slightly outward to make a stable base.

Another way to increase stability is to add **support structures** like **buttresses** (external supports on the sides of walls, seen in old cathedrals) or **guy wires** (cables anchored to the ground, used for radio towers). These prevent tipping by bracing the structure.

In this section, we discuss balance and how forces like wind or uneven loads can tip things. For example, a strong wind is a force that could push a structure. A stable structure (like a well-built tower) will resist the wind by having a good base and perhaps a flexible design that sways but doesn't break. We can demonstrate with an example: two identical cardboard cut-out figures, one with a wide base and one with a narrow base – the one with the wide base stands more firmly.

(Diagram: Two simple tower drawings – one with a narrow base and one with a wide base. An arrow labeled “wind” pushes both. The narrow-base tower is tipping, the wide-base tower stays upright. Caption: “A wider base makes a structure more stable in wind.”)

The idea of **center of gravity** can be introduced in simple terms: it's the point where an object's weight is balanced. If that point is above the base, the object stands; if it moves outside the base, the object tips. A low, central center of gravity (like a weeble toy or a broad-bottom object) aids stability. We'll keep it conceptual for Grade 3 (no calculations, just the concept of balance).

Section 4: Natural vs. Built Structures

Humans aren't the only ones making structures – animals do too, and nature itself creates structures. In this section, we compare **natural structures** and **human-made (built) structures** and see what we can learn from each.

Natural Structures: These include things not made by humans. Examples: honeycombs built by bees (hexagonal wax cells – very strong shape for storing honey), termite mounds (tall mud structures with ventilation – like natural “skyscrapers” for termites), beaver dams (which block water with branches and mud), bird nests (woven twigs and grass to hold eggs safely), and even rock formations or caves can be considered structures formed by nature. Natural structures often use available materials (twigs, mud, etc.) and are optimized for function (a nest is bowl-shaped to hold eggs, a spider web has radial design to catch prey).

We examine one or two case studies: e.g. **a beaver dam vs a human dam** – both slow water flow, built from different materials (wood vs concrete) but both need to be strong enough to hold back water. Or **a bird's nest vs a human house** – both are shelters, one is made of twigs/mud and shaped as a bowl in a tree, the other of wood/brick with walls and a roof. What materials and designs did each “builder” use and why? Students learn that the principles of strength and stability apply broadly: a bird's nest is often in a stable location (fork of branches) and has to support the weight of eggs/bird; its circular shape and woven structure give it strength.

Human-Made Structures: We can highlight how humans sometimes imitate nature. For instance, the honeycomb's hexagon pattern is used in engineering because it's very material-efficient and strong (e.g., hexagon grids in airplane wings or packaging materials). The concept of an arch can be related to how caves arch, or how our foot has an arch for support. Velcro was invented by imitating burs (a plant seed with hooks). This introduces a bit of *biomimicry* – learning from nature's structures to improve our own designs ¹⁸.

By comparing natural and built structures, students appreciate that the same structural concepts (shapes, materials, stability) are universal. It also fosters respect for animal builders – a tiny spider can make a web that's both delicate and remarkably strong for its size!

(Diagram: Two pictures side by side – one of a bird's nest in a tree (natural structure), one of a human-built house. Arrows point to the nest's features (round shape, soft lining, placement in tree forks) and the house's features (roof, walls, foundation). Caption: "Natural and human-built structures both have to be strong and stable for their purpose.")

Section 5: Why Do Structures Matter?

Structures are everywhere in our daily lives, and they matter for both safety and convenience. This final section ties the concept of strong, stable structures to real-world importance and a bit of social/environmental context:

- **Safety:** Buildings, bridges, and playground equipment must be strong and stable so they don't collapse and hurt people. Engineers must design structures to handle more than the expected load (a concept called safety factor). For example, a bridge is designed to hold many cars at once, plus extra weight like wind or snow. If structures are not built well, accidents can happen. We can mention historical examples in simple terms (e.g., "Once, a long time ago, a bridge that wasn't well designed fell down in a strong wind"). This highlights why understanding structure principles is important.
- **Function:** Structures make it possible to span distances (bridges let us cross rivers), to build upward (tall buildings save space on ground by going up), and to create shelter (houses protect us from weather). A strong structure does its job for a long time. A weak structure might break or wear out quickly. We rely on structures for transportation (roads and bridges), for homes and schools, and even small structures like a shelf that holds books or a tent frame for camping. Each is important in its context.
- **Environmental Impact:** We briefly touch on how building structures uses resources (wood from trees, metals from mining, etc.) and why engineers also consider the environment. Strong, durable structures might be better than weak ones that need frequent replacement (less waste). On the other hand, building very large structures can have environmental effects (e.g., a big dam can change a river ecosystem). In Grade 3 terms, we simply introduce the idea that there's a relationship between structures and the environment ¹⁹ ²⁰. For example, "Using stronger materials can let us build bridges that last 100 years, but we should also think about where those materials come from and how building affects nature." Also, maintaining structures (like repairing bridges or houses) is important for safety and sustainability.

By understanding structures, students can also start to do simple design and building on their own. This could lead into an activity like "design and build your own strong and stable structure that serves a purpose" ²¹ – which might be making a small model (e.g., a paper bridge that can hold some coins, or a tower of straws). It's both fun and educational, applying the concepts learned.

Review Questions (Structures Unit)

1. What is a structure?
2. Give one example of a natural structure and one example of a human-made structure.
3. Why are triangles often used in structures?
4. What can make a structure more stable and less likely to tip over?

5. Name a material that is very strong for building. Why is it used?

These questions reinforce the basics: definition and examples of structures (Q1, Q2), key concept of strong shape (Q3), stability concept (Q4), and material choice (Q5). For instance, a good answer to Q3 would be: "Triangles are used because they hold their shape and distribute force evenly, making structures rigid ¹⁵ ." For Q4, students might answer: "Having a wide base makes a structure more stable so it won't tip easily ¹⁷ ." Discussing these answers with students can lead to deeper understanding (e.g., ask them to demonstrate a wide base vs narrow base with objects).

Glossary (Structures Unit)

- **Balance:** A state where a structure's weight is evenly distributed so it stays upright and steady. (Related to stability – a balanced structure doesn't tip over easily.)
- **Compression:** A force that squeezes or presses something together. For example, the weight of a building puts compression force on the columns holding it up.
- **Force:** In physics, a push or a pull on an object ²² . Forces act on structures (like wind pushing a tower, or gravity pulling down on a bridge).
- **Foundation:** The base or bottom part of a structure that sits on the ground, often spread out and made of strong material, which supports the structure's weight. (e.g., the concrete foundation of a house).
- **Load:** The weight or force that a structure supports. There are **live loads** (changing weights like people, cars, wind) and **dead loads** (the structure's own weight). A strong structure can carry its intended load without failing.
- **Material:** What a structure or its parts are made of (wood, metal, plastic, etc.). Different materials have different properties like strength, flexibility, or weight. **Strong materials** like steel or stone help structures support heavy loads ²³ ¹⁴ .
- **Stability:** The quality of a structure that describes how steady it is. A stable structure remains upright and resists tipping or collapsing. **Structures with low center of gravity and wide support** tend to have good stability ¹⁷ .
- **Strength:** The ability of a structure or material to withstand forces without breaking. A strong structure can support a large load. For example, a thick steel bar has more strength than a thin wooden stick.
- **Structure:** A supporting framework with a definite size/shape that holds a load or performs a function ¹² . Structures can be as small as a chair or as large as a bridge – or even part of a living thing (like a skeleton).
- **Triangle:** A three-sided polygon. In structures, triangles are used because they are very rigid and **do not easily change shape under pressure** ¹⁵ (making structures stronger and more stable). Triangles are found in roof trusses, bridge supports, bicycle frames, etc.

(This glossary covers key terms related to structures and mechanisms. It will help students recall the meaning of terms they encountered, like "load" or "stability.")

Booklet 3: Understanding Matter and Energy – Forces and Motion

Title Page & Parent Note: *Forces and Motion*

Note to Parents/Caregivers: This booklet is aligned with the Grade 3 Ontario curriculum strand

“Understanding Matter and Energy: Forces and Motion.” It introduces the concept of forces (pushes and pulls) and how they cause movement. Your child will learn about different types of forces (including contact forces like friction and non-contact forces like gravity) ²⁴, and see examples in daily life. The booklet is organized into short sections with simple language and everyday examples to make these physics concepts accessible. We encourage you to do little experiments at home (for example, feel friction by rubbing hands or see gravity by dropping a ball) to reinforce these ideas. The review questions at the end can guide a fun discussion – you might be surprised by how much your child notices forces around them!

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Section 1: What Is a Force?

In science, **force** means a push or a pull on an object ²². Whenever you push something (like closing a door) or pull something (like tugging a rope), you are using force. Forces can make objects **move, stop, speed up, slow down, or change direction**. For example, kicking a soccer ball (a push) makes it move forward; catching that ball applies force to stop it.

Force is not something we can see directly, but we see its effects. If an object is at rest, a force can get it moving. If it's already moving, a force can change its speed or direction. If multiple forces act at once, they can combine. For instance, if two people push a toy car from opposite sides with equal force, the car won't move because the forces cancel out.

Forces are measured in a unit called **newtons (N)** in physics, but in Grade 3 we focus more on the ideas than the measurements. We will use everyday language like a “strong” or “weak” force. A strong force causes a bigger change in motion than a gentle force. (Think of a gentle push versus a hard shove on a swing – the hard shove sends it much higher!)

Key idea: **No motion happens without a force**. If you see something move or change how it's moving, you can suspect a force is acting on it. Even things that seem to move on their own, like a rolling ball slowing down, involve forces (the ball slows because of friction – more on that later).

*(Diagram: A hand pushing a toy car (with an arrow showing push), and a rope pulling a sled (arrow showing pull).
Caption: “A force is a push or a pull.”)*

Section 2: Contact Forces (Pushes, Pulls, and Friction)

A **contact force** is a force that happens through direct contact – one object touches another to push or pull it ²⁵. Most forces we notice daily are contact forces. Here are some important examples:

- **Applied Push/Pull:** Anytime you directly push or pull an object (open a drawer, push a shopping cart, pull a wagon), that is a contact force. Your muscles provide the force through contact. Likewise, a collision is a contact force – when a moving baseball hits a bat, the bat applies a force to the ball on contact, sending it flying.
- **Friction:** Friction is a contact force that **happens when two surfaces rub against each other** ²⁶. Friction always acts **opposite to the direction of motion**, trying to slow things down. For example, if you slide a book across a table, it eventually stops because friction between the book and table (and air resistance) oppose its motion. Friction can be felt when you rub your hands together – they get warm because friction is converting motion into heat. Without friction, things would slide forever (imagine an ice rink or space where there's very little friction). Friction is useful – it's why we don't slip when we walk (our shoes grip the ground) and how car brakes can stop wheels. But friction can also be something we want to reduce in machines to help parts move smoothly (that's why we use lubricants like oil).
- **Elastic/Spring Force:** When you stretch or compress something elastic (like a spring or a rubber band), it wants to return to its original shape. That “pulling back” force is a contact force within the stretched material. For example, a stretched rubber band can shoot a paper across the room – the force comes from the band snapping back.
- **Buoyancy:** When something is in water, water pushes up on it – this is a contact force from the water (we call it buoyant force). This is why heavy logs can float; the water's upward push counteracts gravity's pull. We mention this as another type of contact force (liquid pushing on an object), though details of buoyancy can be left for higher grades.

Students should understand that any force that requires touching (kicking, dragging, wind blowing on a kite – wind is air in contact) is a contact force. They should also see how friction plays a role all around: why do we have treads on tires (for friction/grip), why smooth ice is slippery (less friction), etc.

(Diagram: Illustration of friction – e.g., a box being pushed across a rough floor, with an arrow showing push forward and a smaller arrow showing friction backward. Possibly magnify to show rough surfaces touching.)

Section 3: Non-Contact Forces (Gravity and Magnetism)

Not all forces require direct touch. **Non-contact forces** can act at a distance, without objects physically touching ²⁷. This might seem magical, but you experience it every day through gravity and magnetism.

- **Gravity:** Gravity is a force of attraction between objects that have mass. In everyday life, **gravity is the force that pulls objects toward the Earth** ²⁶. This is why when you jump, you come back down, and why things fall when dropped. We often talk about gravity as “making things fall,” but it's actually a constant pull. The Earth is so large that its gravity is strong for us; it **pulls everything downwards toward the center of the Earth**. That's why “down” is the same direction as gravity's

pull. Gravity gives us weight – it’s the reason we stay on the ground. If you’ve seen videos of astronauts on the Moon, they kind of float and hop slowly – that’s because the Moon’s gravity is weaker (the Moon is smaller than Earth). In this booklet, we keep it simple: **gravity pulls objects toward Earth**, and it’s always acting on us (even when we’re not moving, like when standing, gravity is pulling us down and the floor is pushing up against our feet).

- **Magnetism:** Have you played with magnets on a fridge? Magnets can **pull (attract)** or **push (repel)** each other **without touching**, as long as they are close ²⁶. Also, a magnet can pull certain metal objects (like iron nails or paperclips) toward it. This is a non-contact force. If you slowly move a magnet closer to a paperclip, at some point the paperclip jumps to the magnet – that’s the magnetic force at work even before contact. Magnets have two poles (north and south); opposite poles attract, like poles repel. We won’t go deep into that in Grade 3, but we will show that magnetism can make things move without touch. An example activity: moving a toy car by holding a magnet near a paperclip attached to the car – the car follows the magnet “like magic”. It’s really the magnetic force pulling the paperclip.
- **Static Electricity:** This is another non-contact force, related to tiny electric charges. If you rub a balloon on your hair and then bring it near small pieces of paper, the paper bits jump to the balloon without touching it. That’s static electric attraction. We include this as a fun fact, though details of positive/negative charges might be above Grade 3. Students just see it as another example of a force that can act through invisible fields.

Non-contact forces often seem invisible because you can’t see a physical push, but they are always present. **Earth’s gravity** is one we can’t escape – it’s why things have weight. **Magnetic forces** are used in many technologies (motors, compasses). Understanding them starts with simple observation: “I didn’t touch it, but it moved!”

(Diagram: Picture of an apple falling from a tree (gravity acting), and a magnet attracting a nail with a gap between them (magnetic force acting through space).)

Section 4: Forces and Movement

Forces cause motion, so in this section we explore how different forces result in different kinds of movement or changes in movement. We also touch on the idea that movement can be described by direction and speed, and forces can change those.

Starting and Stopping Motion: If something is still, a force is needed to start it moving. For example, a soccer ball on grass won’t move until someone kicks (applies force to) it. Once moving, if no other forces act (like in space), it would keep moving. But on Earth, **friction and gravity** usually act to eventually stop motion. That ball on grass slows down and stops because friction from grass (and a bit of air resistance) are forces opposite its motion. To stop a moving object quickly, an opposing force must be applied (like catching a ball or hitting the brakes on a bike – the brake pads create friction on the wheel to slow it).

Changing Speed (Acceleration/Deceleration): Pushing harder (more force) makes an object speed up more. For instance, a gentle push on a swing moves it slowly, a big push makes it go higher and faster. Going downhill on a bike, gravity pulls you forward faster (until friction or air resistance balances it or you brake). Going uphill, gravity is against you, so you slow down unless you pedal harder (apply more force).

This introduces that multiple forces can compete – if forces in one direction are bigger than the other, motion changes in that direction.

Changing Direction: Forces can turn an object. For example, when you swing a ball on a string around you, the pull from the string (centripetal force) constantly changes the ball's direction (making it go in a circle). In simpler terms, think of a rolling toy car that hits a wall – the wall exerts a force that changes the car's direction (it bounces off). Or when playing billiards/pool, hitting the ball at an angle makes it change direction. In daily life, steering a bike involves friction and applied force to change direction.

Multiple Forces: It's worth noting that usually many forces act at once. A book on a table has gravity pulling it down, but it doesn't move because the table pushes up (balancing the force). A flying kite has gravity down, tension in the string, and wind pushing it up – when balanced right, it stays up. We keep it conceptual: balanced forces = no change in motion; unbalanced forces = change in motion ²⁸ (this foreshadows Newton's laws in very basic form).

We also mention simple machines (though that might be more Grade 4 in old curriculum, the new curriculum's STEM skills might integrate it). For context: **simple machines** like levers, inclined planes, wheels, etc., help us use forces more easily (e.g., a ramp allows a weaker force over a longer distance to lift something heavy). But if that's beyond scope, we might skip details and just say humans have tools to make moving things easier.

(Diagram: A sequence showing a boy on a skateboard: first he's still, then he pushes off the ground (force) and starts moving, then he puts his foot down to brake (friction force) and stops. Arrows indicate force directions.)

Section 5: Forces in Our Daily Lives

This section connects the concepts to everyday experiences and technology, to show why understanding forces is useful.

At Home: Many household activities involve forces. Opening a jar = applying torque (a twisting force). Using scissors = applying force at the handles that gets concentrated at the blades to cut (a simple machine lever action). Sitting on a chair = your weight (gravity force) on the chair, which must be strong enough to hold you. Even flipping a light switch or typing on a keyboard involves small forces from your fingers.

Play and Sports: Kicking, throwing, hitting balls – all are forces applied. Think about different sports: in soccer, you apply a force by kicking; in baseball, a bat transfers force to the ball; in basketball, you apply an upward force to shoot, and gravity pulls the ball down through the hoop. Riding a bicycle involves forces – you push on pedals (force to move forward), friction on tires gives grip, brakes use friction to slow down, and gravity helps when going downhill but works against you uphill. **A tug-of-war** game is a great example: two teams pull on a rope. Each team applies force; if forces are equal, the rope doesn't move. If one team pulls harder (greater force), they pull the other team toward them ²⁹.

In Nature: Forces are at work in nature too. Wind is air in motion applying force; it can scatter seeds or even knock down trees in a storm. Water flowing in a river exerts force on soil, causing erosion (wearing away banks) ³⁰. Plants apply forces as they grow – a root can exert force to crack a sidewalk over time! Animals use forces: birds push air with wings to fly, fish push water with fins to swim. Gravity shapes natural events – like water flowing downhill or rain falling.

Technology and Transportation: Cars, buses, and trains move because engines apply forces (through wheels pushing on the road or rails). Seatbelts in cars are there because if the car stops suddenly, your body would keep moving (inertia) and the seatbelt applies a force to hold you safely. Airplanes generate lift (an upward force) with their wings to counteract gravity so they can fly. Magnets are used in things like fridge door seals (to keep it closed via magnetic force) and in electric motors (magnetic forces cause motion inside the motor). Even our Earth's orbit around the Sun is due to gravity – a balance of motion and the Sun's gravitational pull keeps Earth moving around the Sun.

By recognizing these examples, students can see that forces and motion are not just abstract ideas – they are behind **many everyday phenomena and inventions**. Understanding them helps in designing safe structures (e.g., bridges must withstand wind forces) and in using tools/vehicles properly (e.g., knowing you need more force to pedal uphill).

We might end with a note encouraging kids to identify forces around them: “Next time you open a door, think about whether you’re pushing or pulling. When you drop something, remember gravity is at work. Science is everywhere!”

Review Questions (Forces and Motion Unit)

1. Define “force” in your own words.
2. What is the difference between a contact force and a non-contact force?
3. Give an example of a contact force and an example of a non-contact force.
4. What force causes a ball to fall back down after you throw it up in the air?
5. How does friction affect the motion of objects?

These questions reinforce key concepts. For example, a correct answer to Q2: “A contact force requires touching (like push, pull, friction), while a non-contact force acts at a distance (like gravity or magnetism) ²⁵ ²⁷.” For Q4, the answer is **gravity** ²⁶. For Q5, an answer: “Friction is a contact force that slows objects down when two surfaces rub together ²⁶.” Discussing everyday examples in answers (like friction when biking or gravity when something drops) can be encouraged.

Glossary (Forces and Motion Unit)

- **Force:** A push or a pull on an object that can change its motion ²². (E.g., kicking a ball exerts a force that makes it move.)
- **Contact Force:** A force that acts between objects that are touching ²⁵. Examples: pushing a swing, friction between surfaces.
- **Non-Contact Force:** A force that acts at a distance, without direct touch ²⁷. Examples: gravity pulling you down, a magnet attracting iron.
- **Friction:** A contact force that occurs when two surfaces rub against each other, resisting motion ²⁶. Friction can slow down or stop moving objects (e.g., brakes on a bike create friction to stop the wheels).
- **Gravity:** The non-contact force by which Earth (or any large mass) **pulls objects toward its center** ²⁶. Gravity gives objects weight and makes thrown objects fall back down.
- **Magnetism:** A non-contact force where certain metal objects or magnets exert push or pull forces on each other ²⁹. (E.g., a magnet can pull a paperclip without touching it.)

- **Motion:** Movement; a change in position of an object over time. Forces cause changes in motion (starting, stopping, speeding up, slowing down, or turning).
- **Balanced Forces:** Forces that are equal in size and opposite in direction, canceling each other out so that they do not cause a change in motion. (E.g., if two people pull a rope equally hard in opposite directions, the rope doesn't move.)
- **Unbalanced Forces:** When forces on an object are not equal, causing a change in the object's motion (it will accelerate in the direction of the net force).
- **Inertia:** (Advanced term, optional) The tendency of an object to resist changes in its motion. Objects at rest stay at rest, objects in motion stay in motion unless a force acts on them. *(This term might be introduced if discussing why seatbelts are needed – because of inertia.)*

(This glossary provides students with definitions of important terms related to forces and motion. It can be used for reference when completing exercises or discussing the topic.)

Booklet 4: Understanding Earth and Space Systems – Soils in the Environment

Title Page & Parent Note: *Soils in the Environment*

Note to Parents/Caregivers: This booklet is based on the Grade 3 Ontario curriculum strand “**Understanding Earth and Space Systems: Soils in the Environment.**” It explores what soil is made of, the different types of soil, and why soil is so important to living things and the environment. We'll also look at issues like soil erosion and ways we can care for soil. The booklet includes diagrams (in black-and-white outline form) that you can discuss or even have your child color for fun. By working through the sections and questions, students will learn how soil supports plant life and how we can be good stewards of this often overlooked resource. Feel free to dig in (literally, perhaps by examining some garden soil together)!

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Section 1: What Is Soil Made Of?

Soil is not just “dirt” – it's a rich mixture of many things. **Soil is made up of tiny bits of rocks (minerals), decayed plants and animals (organic matter), water, and air, plus lots of living organisms** ³¹. Let's break that down:

- **Rock particles:** These are small pieces of rocks that have been broken down over a long time (through weathering by wind, water, ice, etc.). The pieces can be different sizes – from coarse sand grains to very fine powdery clay. These minerals provide nutrients like calcium, potassium, and iron that plants need.

- **Organic matter:** This means material that comes from living things. In soil, organic matter is mostly **humus** – the dark, rich bits of **decayed leaves, wood, and dead animals** that have broken down. Humus is like a natural fertilizer; it holds nutrients that plants can use and helps soil hold water. It's also why good garden soil looks dark and feels crumbly – lots of humus! **Earthworms, bacteria, and fungi** in the soil help break down dead matter to form humus ³² ³³ .
- **Water:** Soil holds moisture in the spaces between particles. Plants' roots drink from this water. The amount of water soil can hold depends on the soil type (as we'll see in the next section). If you've ever picked up damp soil, that wetness is soil water. Too much water (waterlogged soil) isn't good for most plants because roots also need air.
- **Air:** It might be surprising, but about 25% of healthy soil is air! Tiny pockets of air in soil (in between the particles) are crucial. Roots need air (oxygen) to "breathe," and many soil organisms do too. A soil that's compacted (squeezed with no air spaces) is hard for plants to grow in.
- **Living creatures:** Soil is teeming with life – it's an ecosystem. There are obvious ones like earthworms, ants, beetles, and grubs. Then there are microscopic ones: bacteria, fungi, protozoa, tiny insects, etc. Earthworms are especially important: as they tunnel, they mix the soil and make air spaces; their waste also enriches the soil. A single shovel of garden soil can have billions of microorganisms! These creatures help recycle nutrients by decomposing organic matter.

(Diagram: A cross-section of soil layers – top layer dark with humus and maybe a worm, middle with mix of mineral particles, lower showing rocks – with labels for organic matter, mineral particles, air, water, living organisms.)

The **formation of soil** is slow – rocks break down over hundreds, even thousands of years to make new soil, and organic matter accumulates slowly as plants and animals die and decay. In fact, **it can take more than 500 years to form just an inch of topsoil** ³⁴ ! This is why soil is considered a non-renewable resource in human time scales – once lost, it's not quickly replaced. Understanding what soil is made of helps us appreciate it and use it wisely (for farming, gardening, building, etc.).

Section 2: Types of Soil (Sand, Silt, Clay, Loam)

Not all soils are the same. If you've been to a beach, you've felt **sandy soil** – it's loose, gritty, and doesn't hold water. In contrast, if you've played with clay for pottery, **clayey soil** is heavy, sticky when wet, and holds water too well (puddles). There's also **silt**, which is a fine, powdery soil (often found as mud or river sediment). And a mix of all three plus organic matter makes **loam**, which is generally the best for plants.

Let's describe each main soil type and its characteristics:

- **Sand:** Sandy soil has the largest particles. You can see individual grains and feel them (gritty). Because the grains are big, there are large spaces between them. This means water drains through quickly (sand can't hold water well) and it dries out fast. Air can get in easily, which is good, but nutrients tend to wash out with the water. Plants growing in sand may need frequent watering and fertilizing. However, sand doesn't get muddy – it's easy to dig and roots can penetrate it easily. Think of cactus in the desert – desert soil is very sandy and doesn't hold much water, and those plants are adapted to that.
- **Clay:** Clay has the tiniest particles – so small you can't see them individually without a microscope. When clay is wet, it feels sticky and can be molded (just like art clay). The tiny particles pack tightly,

leaving little space for air or water movement. So clay soil holds water a long time (water doesn't drain, leading to puddles or soggy ground) ³⁵ . When it dries, it can become very hard and even crack. Clay soils are often rich in nutrients (the particles are good at holding onto nutrients), but the challenge is plant roots can suffocate or rot in heavy clay because of poor aeration and drainage. Farmers often find clay soil difficult to plow when wet (it's heavy) and hard as brick when dry. Some plants, like rice, can tolerate clay (rice paddies are flooded fields in clay soil!).

- **Silt:** Silt particles are medium-sized – smaller than sand, larger than clay. Silty soil feels silky or floury when dry (imagine fine dust) and smooth mud when wet. It holds water better than sand but not as much as clay – can be thought of as intermediate. Silt is often found in river valleys (left by floods). It's fertile and good for farming if drained well, but pure silt can compact easily. (We might not emphasize silt as much at Grade 3, but it's good to mention as part of loam.)
- **Loam:** Loam is a word for a soil that is a **balance of sand, silt, and clay, plus organic matter**. It's considered ideal for gardening and farming. **Loam soils hold on to nutrients and water effectively, but also drain enough and have air for roots** ³⁶ ³⁷ . They are usually dark, soft, and crumbly – easy to dig. An approximate “perfect” loam composition is 40% sand, 40% silt, 20% clay, with a good amount of humus ³⁷ . Most topsoil sold in garden centers is loam. Plants generally love loamy soil because it provides a balanced environment (not too dry like sand, not waterlogged like clay).

How can one tell soil type? A simple test: **the feel test** – rub moist soil between your fingers. Sandy soil feels gritty; silty soil feels smooth like flour; clay soil feels sticky. Another is **the ribbon test** – roll moist soil into a ball and then a “snake” or ribbon. Sand won't hold shape (falls apart). Clay will form a long ribbon that holds together (and a ball that doesn't break easily). Loam will form a short ribbon that breaks apart – a mix of characteristics ³⁸ ³⁷ .

Each soil type has its uses. For example, for building foundations, very sandy soil might not be great because it shifts, but a mix with some clay is more stable. For plant growth, loam is best. This section might include a fun activity suggestion: get small jars, put some soil, add water, shake and let settle – sand, silt, clay will settle in layers (heaviest sand at bottom, silt in middle, clay on top) to see the components.

(Diagram: Three jars or illustrations showing sand (large particles, water quickly draining through), clay (tiny particles, water pooling), and loam (mix of particle sizes with water holding but also draining). Or a soil texture triangle – but that might be too advanced. A simpler diagram: one panel with “sand – large grains”, one with “clay – tiny particles”, one “loam – mix”, with water arrows showing drainage differences.)

Section 3: Soil and Plants: Why Soil Is Important

Soil is often called “the foundation of life on land” because **most plants need soil to grow**, and plants support other life (providing food and oxygen). This section discusses why soil is so critical, especially focusing on plants and ecosystems:

- **Anchor and Home for Plants:** Soil gives plants a place to anchor their roots. A plant's roots spread through soil, which holds the plant upright. Without soil (or something to root in), many plants would topple or be unable to grow tall. Some plants can grow in water (hydroponics) or on rocks, but generally, soil is their home.

- **Nutrients and Water:** Soil provides **nutrients** (like nitrogen, phosphorus, potassium) that plants absorb through their roots. These nutrients come from the minerals in rock particles and from decomposed organic matter (humus). We often add fertilizer to soil to replenish nutrients for crops. Soil also **stores water** for plants after it rains. The **best soils (loams)** hold enough water for plants but also drain excess so roots can breathe ³⁷. Poor soils either hold too little water (sands) or too much (clays), affecting plant health. Healthy soil acts like a sponge – soaking up rainfall and slowly releasing it to plant roots.
- **Habitat for organisms:** Soil is an ecosystem. Many creatures live in the soil (as mentioned earlier). These organisms help plants: for example, some bacteria convert nitrogen from air into forms plants can use (nitrogen fixation); worms aerate the soil; fungi (mycorrhizae) attach to roots and help them absorb water/nutrients. In turn, plants feed soil organisms by shedding leaves or through root chemicals. It's a busy, mutual relationship.
- **Support for Food Webs:** Because soil grows plants, and plants feed herbivores, which feed carnivores, etc., the quality of soil can affect entire food chains. Rich soil = healthy plants = more food for animals. Poor soil (lacking nutrients or water) can mean sparse plant life, affecting animals. We can mention how different ecosystems have different soils – e.g., a rainforest has a thin soil but rapid recycling of nutrients, a prairie has deep rich topsoil from grasses, a desert has sandy soil that supports only adapted plants.
- **Infrastructure and Water Filtering:** Beyond biology, soil is important as the ground we build on. Engineers study soil before building to ensure it can support structures. Also, soil filters water – when rain seeps through soil, the soil can trap pollutants, resulting in cleaner groundwater. Wetland soils, for example, can clean water naturally.

It's worth noting that **soil quality** matters for agriculture: places with very fertile soil (like river deltas or volcanic valleys) have supported civilizations. When soil is depleted or eroded, it can lead to problems like poor crop yields or even famine. So caring for soil is a big part of sustainable living, which leads into later sections.

Section 4: Erosion: When Soil Goes Missing

Soil erosion is the process of soil being carried away by wind or water ³⁰. While some soil movement is natural (for example, river floods deposit silt elsewhere), **too much erosion is a problem** because it removes the topsoil, which is the most fertile layer.

Causes of erosion:

- **Water:** Rainfall can wash soil away, especially on slopes. You might have seen muddy water running off fields or gardens during heavy rain – that water is carrying soil with it. Rivers can erode their banks, and ocean waves erode beaches and cliffs. When trees or plants that held soil in place are removed, water can wash the soil much more easily (like if you clear a hillside of vegetation, the next rain might cause a landslide or lots of runoff).
- **Wind:** In dry areas, wind can pick up fine soil (dust) and blow it far away. History example (if teacher chooses): the Dust Bowl of the 1930s, where poor farming left fields bare and wind carried away huge amounts of topsoil in dust storms.
- **Gravity:** On steep slopes, gravity can pull soil downhill in landslides or mudslides, especially when the soil

is wet and heavy (water loosens the soil).

- **Human Activity:** When we plow fields, cut down forests, or overgraze land with livestock, we often expose soil and make it easier for wind and water to erode it. Construction can also loosen soil. That's why you often see silt fences around construction sites – to catch eroding soil so it doesn't pollute waterways.

Why erosion is bad: We lose fertile topsoil which plants need. The soil that washes away often ends up in rivers or lakes, causing problems like siltation (filling up waterways, harming fish and water quality). Nutrients from soil can cause water pollution or algae blooms. Also, without soil, land can turn barren (desertification in extreme cases). Farmers then have to work harder (or use more fertilizer) to grow crops on remaining poorer subsoil.

How to observe erosion: Kids may notice after a rain that soil has moved – e.g., little gullies in a dirt path, or mulch washed out of a garden bed. On a small scale, that's erosion. On a large scale, entire riverbanks can collapse.

We will include how plants help prevent erosion. Plant roots hold soil together and plants break the force of raindrops on the ground. A grassy lawn will erode less than bare soil in a storm. Forests on slopes protect against landslides by anchoring soil. So one key to preventing erosion is to keep soil covered with vegetation.

(Diagram: A before-and-after of a hillside: before – with plants, rain falls and only little runoff; after – no plants, rain causes a lot of soil to wash down the hill. Possibly show a river carrying sediment.)

We'll also mention human measures: farmers use techniques like contour plowing (plow along hills in horizontal lines), terracing (making flat steps on hills), planting windbreaks (rows of trees), and cover crops to reduce erosion ³⁹. Construction sites might spray water or use tarps to keep dust down (prevent wind erosion) and use retaining walls or mats to hold soil.

Understanding erosion ties back to forces (from the previous unit – water and wind are forces acting on soil) and to caring for the environment. It's a nice link between science strands.

Section 5: Caring for Soil (Compost and Conservation)

Soil is a valuable resource, and this final section talks about how we can take care of it or improve it. Some key points:

- **Composting:** Making compost is a way to recycle organic waste (like vegetable peels, leaves, grass clippings) into rich humus for soil. **Compost is a mix of rotted organic material that can be added to soil to help plants grow** ⁴⁰. By composting kitchen scraps and garden waste, we reduce garbage and create natural fertilizer. Compost improves soil structure (adds humus to sandy or clay soils as mentioned earlier) ⁴¹ and returns nutrients to the soil. A simple compost bin in the yard can teach kids how worms and microbes break down banana peels into brown soil-like material over weeks or months. Using compost in a garden makes soil dark, rich, and smell earthy – a sign of health. The booklet can include a small “how to compost” box or just an explanation of the concept.
- **Avoiding Pollution:** We need to be careful what goes into the soil. Hazardous chemicals (like pesticides, oil spills, etc.) can contaminate soil, killing the helpful organisms and making it unsafe for

growing food. Even litter (plastic, etc.) doesn't break down and can harm soil life. Encourage keeping soil clean – e.g., disposing of motor oil properly, reducing chemical use in the garden by using organic methods if possible.

- **Preventing Erosion:** As discussed, **keeping plants on soil** is a major way to conserve it. Planting groundcover or trees on slopes, not leaving fields bare (farmers often plant cover crops or use mulch). If you have a garden, using mulch (straw, wood chips) can protect soil from eroding in heavy rain and also keep moisture in. Community efforts like planting trees, restoring wetlands, or creating windbreaks help protect soil at larger scales.
- **Smart Farming Practices:** Mention how farmers rotate crops (different plants each season) so the soil isn't depleted of one kind of nutrient, and some plants (like beans) actually add nutrients (nitrogen) back. Also, not plowing too deeply or frequently can preserve soil structure (techniques like no-till farming keep last year's plant stalks on the field to protect the soil). These ideas show that people are actively working to keep soil healthy for future generations.
- **Soil in our community:** Perhaps encourage students to think of soil around them – a garden, a park, a schoolyard. Is the soil covered with grass or plants? That's good. If they see an area of bare soil, they might notice it turns muddy or erodes – maybe that area could be planted. Even small actions like planting a little garden or not stepping on fragile soil (to avoid compaction and plant damage) can be ways kids help soil.

In summary, caring for soil = adding organic matter (compost), avoiding harmful substances, and preventing erosion. Healthy soil means healthier plants, cleaner water (since soil filters water), and a better environment.

(Diagram: Perhaps a simple compost pile illustration (banana peel, leaves arrow to compost), or showing hands spreading compost in a garden. Or a comparison: a thriving plant in rich soil vs a weak plant in poor soil.)

This section concludes the content by emphasizing that soil, though it's under our feet, deserves as much care as air and water. It literally sustains us through food production and natural ecosystems.

Review Questions (Soils Unit)

1. **What are two things that soil is made of?**
2. **Name the three main types of soil particles. Which one feels gritty, and which one feels sticky?**
3. **Why is loam considered the best soil for plants?**
4. **What is soil erosion, and what can cause it?**
5. **How can we help keep soil healthy and prevent it from being lost?**

These questions cover composition (Q1 might be answered: "Soil is made of rock particles and humus (decayed plants/animals), plus also air and water." ³¹), soil types (Q2: "Sand is gritty, clay is sticky when wet; silt is smooth."), loam (Q3: "Loam has a mix of sand/silt/clay and humus, holding water and nutrients well ³⁶."), erosion (Q4: "Erosion is when soil is carried away by water or wind ³⁰, for example, heavy rain washing soil off a hillside."), and conservation (Q5: e.g., "We can plant grass/trees to hold soil, and add compost to keep it rich, and avoid polluting it."). Discussing these encourages understanding of both facts and practical actions.

Glossary (Soil and Environment Unit)

- **Clay:** A type of soil with very tiny particles. Feels sticky when wet and clumps together. Clay soil holds water a long time and can lack air for plants. (In soil textural terms, clay is the finest particle.)
- **Compost:** Decayed organic matter (like food scraps, leaves) that is added to soil to improve its quality ⁴⁰. Compost is rich in nutrients and helps soil hold water; gardeners often call it “black gold.”
- **Erosion:** The process by which soil (or rock) is worn away and moved from one place to another by wind, water, or ice ³⁰. (Example: rain washing topsoil off a hillside.)
- **Humus:** The dark, crumbly, nutrient-rich part of soil formed from decayed plant and animal material. Humus helps make soil fertile and holds moisture.
- **Loam:** A well-balanced soil type containing a mix of sand, silt, and clay, plus organic matter ³⁶. Loam is ideal for growing plants because it retains water and nutrients but also drains well.
- **Nutrient:** A substance that living things need to grow. Soil nutrients (like nitrogen, phosphorus, potassium) are absorbed by plant roots. Fertile soil has plenty of nutrients, often supplied by humus or fertilizers.
- **Organic Matter:** Material that comes from living organisms. In soil, this refers to dead and decomposed plant parts, manure, etc. that enrich the soil. (Organic = from living things, not to be confused with “organic” meaning no chemicals in food, though related concept.)
- **Sand:** A type of soil with large, gritty particles. Sandy soil drains water very quickly and doesn’t hold nutrients well. It feels coarse to the touch.
- **Silt:** Soil with medium-sized particles, feeling smooth or floury when dry. Silt holds water better than sand but can be compact. Often found as fine sediment in river areas.
- **Topsoil:** The uppermost layer of soil, usually dark and rich in organic matter. Topsoil is where most plant roots grow and where most soil organisms live. It’s the most productive layer of soil for plant growth, and it’s the layer that is lost first to erosion if not protected.

(This glossary provides definitions of key terms about soil. It will help students with vocabulary such as the soil types and concepts like erosion and compost.)

Booklet 5: “Just the Facts” – Grade 3 Science One-Liners

(This is a standalone reference booklet with simple facts spanning all Grade 3 science strands. It has no specific sections or images – just concise, one-line factual statements that students can read and remember. Each fact is independent, and together they cover a broad range of Grade 3 science tidbits. This booklet can be printed as 20 pages of large text, each page containing a few facts, making it easy for quick reference or review.)

- A frog is an **amphibian** (it can live in water and on land) ⁴².
- All living things need **water** to survive ⁴.
- Plants make their own food using **sunlight** (through a process called photosynthesis) ⁶.
- Green plants give off **oxygen** that animals and people need to breathe ¹¹.
- The **Sun is a star**, and it is the center of our solar system ⁴³.
- The Sun’s energy supports all life on Earth (it helps plants grow, which feed animals).
- **Roots** hold a plant in the ground and take up water and nutrients.
- **Leaves** use sunlight to help plants make food.
- **Flowers** can turn into fruits that contain seeds for new plants.

- A **structure** is a framework that has a shape and can support a load ¹² .
- **Triangles** are often used in buildings because they are a very strong shape ¹⁵ .
- Structures are more **stable** (steady) when they have a wide base ¹⁷ .
- Bridges, houses, and nests are all **structures** (some made by humans, some by animals).
- A **force** is a push or a pull that can make something move ²² .
- **Gravity** pulls objects toward the Earth's ground ²⁶ .
- **Friction** is a force that slows things down when two surfaces rub together ²⁶ .
- Magnets can **attract** (pull) or **repel** (push away) certain metals without touching them ²⁶ .
- When you throw a ball up, it comes down because of gravity.
- The rougher a surface, the more friction it produces (e.g., carpet slows a toy car more than a smooth floor).
- Earth is surrounded by an **atmosphere** of air, which we breathe.
- **Soil** is a mix of tiny rocks, decayed plants/animals, water, and air ³¹ .
- **Sand** is a type of soil with large grains that water passes through quickly.
- **Clay** is a type of soil with very small particles; it feels sticky and holds water.
- **Loam** is soil that's great for plants – a balance of sand, silt, clay, plus humus ³⁶ .
- **Erosion** happens when wind or water carries soil away ³⁰ .
- Planting trees or grass can help prevent soil erosion on bare land.
- Earthworms tunnel in soil, helping to mix it and making it healthier for plants.
- Water can exist in three states: solid (ice), liquid (water), and gas (water vapor).
- **Matter** is anything that has mass and takes up space (solids, liquids, gases are states of matter).
- When you mix certain things, you can cause a reaction (e.g., baking soda and vinegar fizz to make a gas).
- Light travels in a straight line until it hits an object.
- A **shadow** is made when an object blocks light.
- Sound is caused by vibrations and travels through air (or other materials) to reach our ears.
- Electricity can flow through a circuit to power a bulb or device.
- A **simple machine** like a ramp or a pulley can help us use forces more easily.
- The Earth rotates once every 24 hours, causing day and night.
- The Earth orbits around the Sun once every year (about 365 days).
- The Moon orbits around the Earth and has phases (new moon, full moon, etc.).
- **Animals** can be grouped by how they are born and live (mammals, birds, reptiles, amphibians, fish).
- Birds have feathers, lay eggs, and most can fly.
- Fish have gills and scales, and live in water.
- **Reptiles** are cold-blooded, have scales, and usually lay eggs (like snakes and lizards).
- **Mammals** are warm-blooded, have hair or fur, and most give birth to live young.
- The **life cycle** of a butterfly: egg → caterpillar (larva) → pupa (chrysalis) → adult butterfly.
- **Habitat** is the natural home of a living thing (e.g., a forest, a pond, a desert).
- **Recycle, Reduce, Reuse** are ways we can help protect Earth's resources and environment.

(Each of these facts is written as a simple sentence. There are no images or questions in this "Just the Facts" booklet – just plain statements that students can read for quick information. Teachers or parents can print these on separate pages or as a list. It spans life science, physical science, and Earth science topics, giving a broad review of Grade 3 science knowledge.)

Sources: The content in these booklets has been informed by the Grade 3 Ontario Science & Technology curriculum and educational resources. For example, the strands and topics are as outlined by TVO Learn for

Grade 3 ⁴⁴ . Specific facts have been cross-checked with reliable sources: plant needs and photosynthesis ³ ⁶ , definitions of structures and stability ¹² ¹⁷ , examples of forces ²⁶ , and components of soil and erosion information ³¹ ³⁰ , among others. These references ensure accuracy and alignment with curriculum expectations while keeping explanations age-appropriate.

¹ ⁹ ¹⁸ ²¹ ³⁹ ⁴⁴ **Grade 3 - Science and Technology | TVO Learn**

<https://tvolearn.com/pages/grade-3-science-and-technology>

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³ **Growth and Changes in Plants - Science and Technology - Ontario.ca**

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⁷ **Plant and Animal Life Cycles | Science Lesson For Kids | Grades 3-5**

<https://www.generationgenius.com/plant-and-animal-life-cycle-lesson-for-kids/>

¹² ¹⁴ ²³ ²⁵ ²⁷ **scitechontario.ca**

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¹⁵ **Triangles: The Strongest Shape - High Touch High Tech**

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¹⁶ **The Secret Strength of Triangles - Saint Louis Science Center**

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¹⁷ **[PDF] -Structures are more stable when they have a wider base ...**

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¹⁹ **[PDF] A Strong Wall - Grade 3 (Strong and Stable Structures)**

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²⁰ **5 Creative Ways to Teach Strong and Stable Structures**

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²² ²⁸ **Physics for Kids: Force**

<https://www.ducksters.com/science/physics/force.php>

²⁴ **Forces and Motion - Science and Technology - Ontario.ca**

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